LabTech Testing Limited

STANDARD TEST PROCEDURE

ASTM D5731
Determination of the Point Load Strength Index of Rock
for "Diametral Test" and "Axial Test"

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REVISION LEDGER

Rev. No.	Section No.	Page(s)	Revision Outline
1	NA	NA	First Issue
2	Cl. 2.1	3	Added exemption of rock specimens in form of blocks or irregular lumps
	Cl. 5.3.4	6	Revised reference to Figure 4
	Cl. 6.3.6	8	Revised reference to Figure 6

1. Introduction

1.1 This test method covers the equipment, field procedures, interpretation of data for the determination of point load strength index of rock.

1.2 The test method is prepared based on ASTM D5731-16 Standard Test Method for Determination of the Point Load Strength Index of Rock and Application to Rock Strength Classifications and ISRM (1985) Suggested Method for Determining Point Load Strength.

2. Scope

2.1 This procedure specifies the method of the guidelines, requirements and procedures for determining the point load strength index of rock for "Diametral Test" and "Axial Test" only. Rock specimens in the form of blocks or irregular lumps could not be tested by this test method.

2.2 This procedure is performed to determine the point load strength index Is(50) of rock specimen that is the ratio of point load strengths on different axes that result in greatest and least values, which is intended to be used for strength classification of rock materials.

2.3 Rock specimens in the form of core either "Diametric Test" or "Axial Test" is tested by application of concentrated load through a pair of truncated, conical platens, until failure occurs by splitting the specimen. Little or no specimen preparation is required.

2.4 This procedure can be performed in the field due to the testing machine is portable and applies to hard rock (compressive strength over 15 MPa) and the values stated in the SI units are to be regarded as standard.

3. Limitations

3.1 This procedure is not suitable for determination of the point load strength index of rock to other standard or document.

4. Testing Equipment

4.1 Point Load Tester

4.1.1 The testing equipment shall be calibrated prior to use.

4.1.2 A point load tester (*Figure 1*) consists of a loading system typically comprised of a loading frame, platens, a measuring system for indicating load, **P** (required to break the specimen) and a means for measuring the distance, **D**, between the two platen contact points. The equipment shall be resistant to shock and vibration so that the accuracy of readings is not adversely affected by repeated testing.

4.1.3 The loading system shall have a loading frame with a platen-to-platen clearance that allows testing of rock specimens in the required size range. Typically, this range is within 30 to 100mm so that an adjustable distance is available to accommodate both small and large specimens.

4.1.4 The loading capacity shall be sufficient to break the largest and strongest specimens to be tested.

4.1.5 The test machine shall be designed and constructed so that it does not permanently distort during repeated applications of the maximum test load and so that the platens remain coaxial within ±0.2 mm throughout testing. No spherical seat or other non-rigid component is permitted in the loading system. Loading system rigidity is essential to avoid slippage when specimens of irregular geometry are tested.

4.1.6 Truncated, conical platens, as shown on *Figure 2*, are to be used. The 60° cone and 5 mm radius spherical platen tip shall meet tangentially. The platens shall be of hard material (Rockwell 58 HRC) such as tungsten carbide or hardened steel so they remain undamaged during testing.

4.1.7 The load measuring system, for example a load cell or a hydraulic pressure gage, that will indicate failure load, **P**, required to break specimen and shall be to a precision of ±5 % or better of full-scale load-measuring system, irrespective of the size and strength of specimen that is tested. Failure is often sudden and a peak load indicator is required so the failure load can be recorded after each test.

4.1.8 The system should be capable of using interchangeable measuring devices in order to be consistent with the estimated strength of rock (point load strength of rock is usually an order of magnitude lower than the compressive strength of rock).

4.1.9 The Distance Measuring System a vernier direct reading scale, should connect to the loading frame for measuring the distance, **D**, between specimen-platen contact points and shall be to an accuracy of ±2 % or better of distance between contact points, irrespective of the size and strength of specimen that is tested. The Distance Measuring System also shall allow a check of the "zero displacement" value when the two platens are in contact and should include a zero adjustment. An instrument such as a caliper or a steel rule is required to measure the width, **W**, (with an accuracy of ±5 %) of specimens for all but the Diametral Test.

4.1.10 Rock samples are grouped on the basis of both rock type and estimated strength. When testing core specimens at least ten specimens are selected. Specimens in the form of core are preferred for a more precise classification.

4.2 Monitoring and Measurement Equipment

- Spherically-truncated
- Vernier calipers
- Steel ruler
- Hydraulic pressure gage

The monitoring and measuring equipment shall be calibrated prior to use.

5. Testing Procedure

5.1 In general, all the processes of Point Load Test shall be carried by the Qualified Operator, supervised and checked by the Approved Signatory.

5.2 Preparation

5.2.1 The customer shall provide the following information before the test is conducted by filling in the Test Order Form for Point Load Test (*Form No. FQS08*) requiring:

- Drillhole no.
- Sample no.

- Depth.

5.2.2 For the test performed in the field, the specimen at their natural water content unless otherwise specified by the customer.

5.2.3 The specimens shall be properly marked and measured.

5.2.4 Marking - The desired test orientation of the specimen shall be indicated by marking lines on the specimen using thin water-proof marker pen. These lines are used for centering the specimen in the point load tester and to ensure proper orientation during testing. These lines may also be used as reference lines for measuring thickness and diameter.

5.2.5 Measuring - Measure the dimension of the specimen at three different places of the specimen, and calculate the average. Measure the length of the specimen.

5.3 For Diametric Test

5.3.1 Core specimens with **Length/Diameter** ratio greater than **1** are suitable for diametric testing.

5.3.2 Specimen should be marked before test with a thin water-proof marker pen along the circumference perpendicular to the core axis and 180∘where the point load will applied. Determine and record the distances of **D** and **L** (shown in *Figure 3*).

5.3.3 Insert a specimen in the point load tester and close the platens to make contact along a core diameter. Ensure that the distance, **L**, between the contact points and the nearest free end is at least 0.5 times the core diameter, **D**, (shown in *Figure 3(a)*), i.e. **L≧0.5D**.

5.3.4 Steadily increase the load such that failure occurs within 10 seconds to 60 seconds and record failure load, **P**. The test should be rejected if the fracture surface passes through only one platen loading point (shown in *Figure 4*).

5.3.5 The procedures in section 5.3.2 – 5.3.4 are repeated for each test specimen of the rock type.

5.4 For Axial Test

5.4.1 Core specimens with **Length/Diameter** ratio (**W**) of **1/3** to **1** are suitable for axial testing (shown in *Figure 3(b)*). Suitable specimens can be obtained by saw cutting

5.4.2 Specimen should be marked before the test with a thin water proof marker pen, indicating the centers of the end faces where the point load will be applied. Record the distance, **D**, between platen contact points (shown in *Figure 3*) as well as record the specimen width, **W**, perpendicular to the loading direction with an accuracy of ±5%.

5.4.3 Insert a specimen in the point load tester and close the platens to make contact along a line perpendicular to the core end faces.

5.4.4 Steadily increase the load such that failure occurs within 10 seconds to 60 seconds and record failure load, **P**. The test should be rejected if the fracture surface passes through only one platen loading point (shown in *Figure 4(e)*).

5.4.5 The procedures in section 5.4.2 – 5.4.4 are repeated for each test specimen of the rock type.

6. Calculation

6.1 Uncorrected Point Load Strength Index - The uncorrected point load strength $\mathbf{I_S}$ is calculated as:

$$\mathbf{I_S = P/D_e^2, MPa}$$

where:

P = Failure Load, N

D_e = Equivalent Core Diameter = D for diametric tests (shown in *Figure 3*), mm and is given by:

D_e^2 = D^2 for cores, mm^2, or

D_e^2 = $4A/\pi$ for axial, block and lump tests, mm^2

where:

A = W*D = minimum cross-sectional area of a plane through the platen contact points (shown in *Figure 3*).

6.2 If significant platen penetration occurs in the test, such as when testing weak sandstone, the value of **D** should be the final value of the separation of the loading points, **D'**. Measurements of core Diameter, **D** or specimen width, **W**, made perpendicular to the line joining the loading points are not affected by this platen penetration and should be retained at the original values. The modified values of $\mathbf{D_e}$ can be calculated from:

$$\mathbf{D_e^2 = D * D'} \text{ for cores} = \mathbf{(4 /\pi) * W * D'} \text{ for other shapes}$$

6.3 Size Correction Factor

6.3.1 $\mathbf{I_s}$ varies as a function of **D** in the diametric test and as a function of $\mathbf{D_e}$ in axial, block and irregular lump tests, so that a size correction must be applied to obtain an unique point load strength value for the rock sample and one that can be used for purposes of rock strength classification.

6.3.2 The size-corrected point load strength index, $\mathbf{I_{s(50)}}$ of a rock specimen is defined as the value of Is that would have been measured by a diametric test with **D = 50 mm**.

6.3.3 When a precise rock classification is essential, the most reliable method of obtaining $\mathbf{I_{s(50)}}$ is to conduct diametric tests at or close to **D = 50mm**. Size correction is then unnecessary. For example, in case of diametric tests on **NX**, core diameter = **54 mm** and size correction to **D = 50 mm** is not necessary. Most point load strength tests are in fact performed using other specimen sizes or shapes. In such case, the size correction described in section 6.3.4 or 6.3.5 must be applied.

6.3.4 The most reliable method of size correction is to test the specimen over a range of **D** or $\mathbf{D_e}$ values and to plot graphically the relation between **P** and $\mathbf{D_e}$. If a log-log plot is used, the relation is a straight line (shown in *Figure 5*). Points that deviate substantially from the straight line may be disregarded (although they should not be deleted). The value of $\mathbf{I_{s(50)}}$ corresponding to $\mathbf{D_e^2 = 2500\ mm^2}$ ($\mathbf{D_e = 50\ mm}$) can be obtained by interpolation and use of the size-corrected point load strength index calculated as shown in 6.3.5.

6.3.5 When neither 6.3.3 nor 6.3.4 is practical (for example when testing single-sized core at a diameter other than 50 mm or if only a few small pieces are available), size correction may be accomplished using the formula:

$$\mathbf{I_{s(50)} = F * I_s}$$

6.3.6 The Size Correction Factor **F** can be obtained from the chart (shown in *Figure 6*) or from the expression:

$$\mathbf{F = (D_e/50)^{0.45}}$$

6.3.7 For tests near the standard 50 mm size, only slight error is introduced by using the approximate expression:

$$\mathbf{F = (D_e/50)^{0.5}}$$

6.3.8 instead of using the procedure outlined in *Figure 5*.

6.4 Mean Value Calculation

6.4.1 Mean values of $\mathbf{I_{s(50)}}$ defined in 6.4.2, are to be used when classifying samples with regard to their point load strength and point load strength anisotropy indices.

6.4.2 The mean value of $\mathbf{I_{s(50)}}$ is to be calculated by deleting the two highest and two lowest values from the ten, or more, valid tests and calculating the mean of the remaining values. If significantly fewer specimens are tested, only the highest and lowest values are to be deleted and the mean calculated from those remaining.

7. Reporting

7.1 After collecting the necessary data and information which is recorded on Worksheet (*From no. FST06-RKT-D5731*), the Determination of Point Load Strength of Rock Test Report (*Report no. RKT-D5731-RT*), would be prepared. The Point Load Test Records and results shall be checked by the Approved Signatory. Besides, the Point Load Test Report shall be endorsed by the Approved Signatory prior to issue.

8. Measurement Uncertainty

8.1 Measurement uncertainty shall be determined for the equipment and procedures used, but will also be affected by parameters, such as grade of rock, diameter of rock and length of rock.

8.2 Therefore, the expanded uncertainty (**U**) of pressure measurement at confidence level 95% with coverage factor of 2 shall be **0.000886 MPa**.

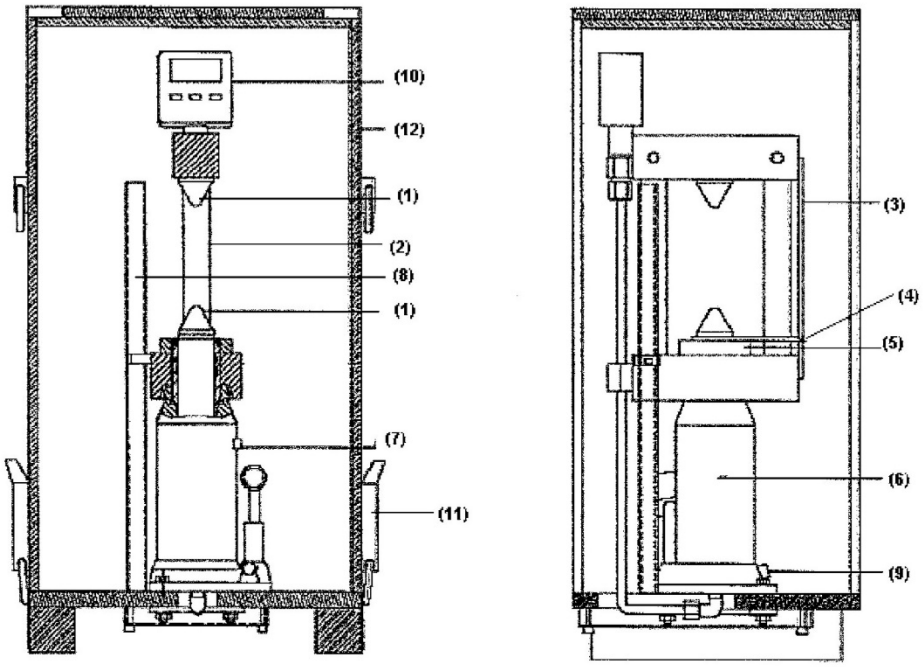
9. Specific Calibration / Verification Requirements

9.1 The specific calibration / verification requirements for equipment of Point Load Test (PLT) are as follows:

9.2

Type of Equipment	**Maximum Period between successive calibration / verification**	**Calibration / verification procedures or guidance documents and equipment requirements**
Load verification	1 Year	By a "competent calibration body" as defined in clause 2.1 of HOKLAS SC-02
Coaxiality of load pointers	6 Months	By use of appropriate reference measuring device
Geometry of the cone	Initial and before use	By use of appropriate reference measuring device
Hardness of the cone	Initial only	By use of appropriate reference measuring device

Figures & Tables



Standard hardened points (1), Fixed crosshead frame (2), Scale (3), Scale pointer (4), Attaching bolt (5), Hydraulic pump body (6), Oil filler cap (7), Pump handle (8), Pressure release valve (9), Digital pressure readout (10), Apparatus protective cover (11, 12)

Figure 1 – Example of a Point Load Tester

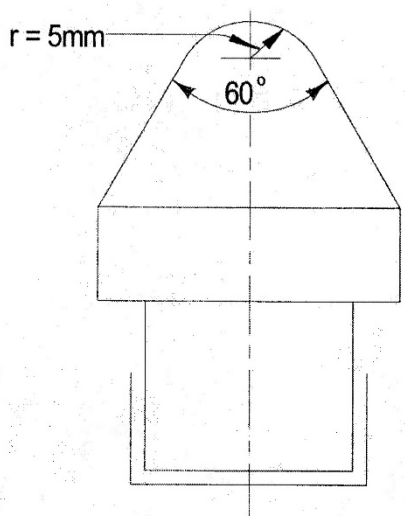


Figure 2 – Platen Dimensions

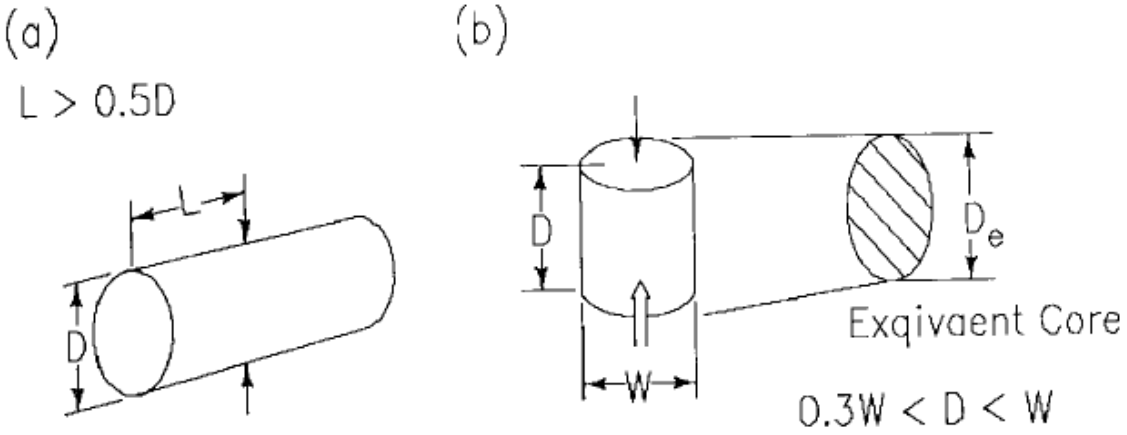


Figure 3 – Load Configurations and Specimen Shape Requirement for (a) Diametral Test, and (b) Axial Test

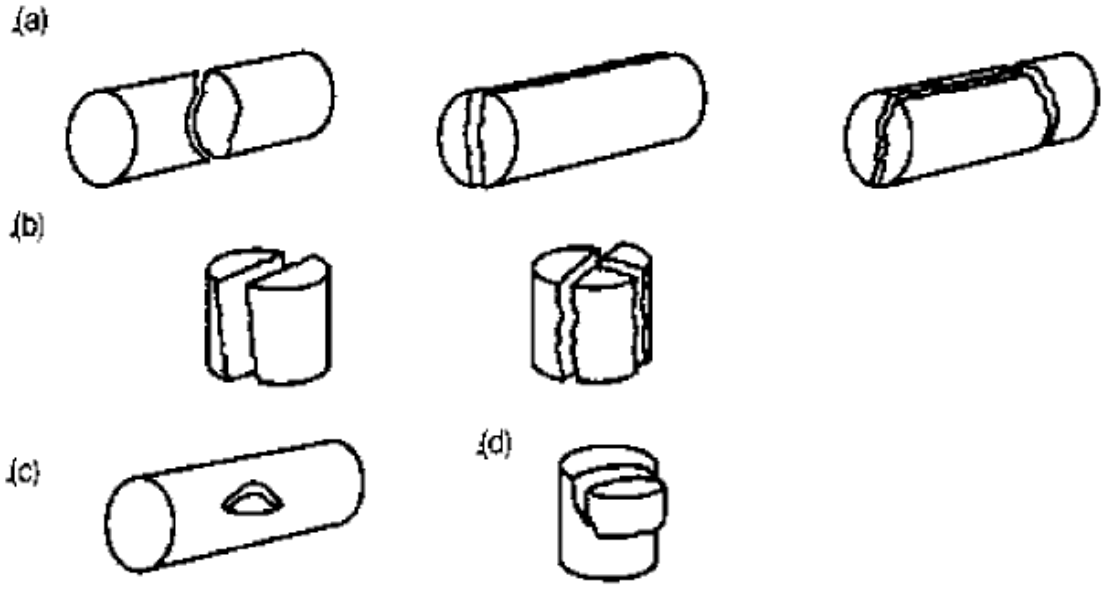


Figure 4 – Typical Modes of Failure for Valid and Invalid Tests

(a) Valid Diametral Tests, (b) Valid Axial Tests, (c) Invalid Core Test and (d) Invalid Axial Test (Point Load Strength Index Test)

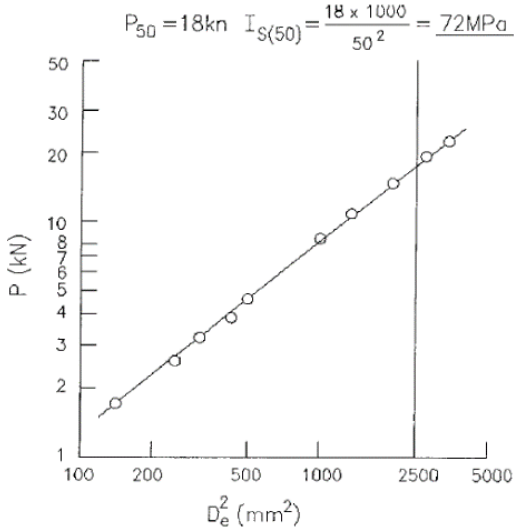


Figure 5 – Procedure for Graphical Determination of $I_{S(50)}$ from a Set of Results at D_e Values other than 50mm

(ISRM Suggested Method)

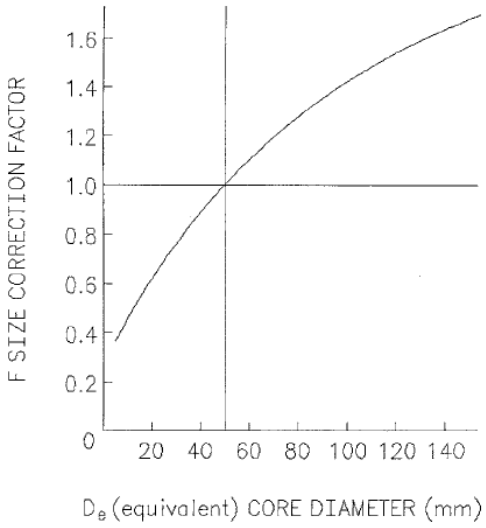


Figure 6 – Size Correction Factor Chart

(ISRM Suggested Methods)

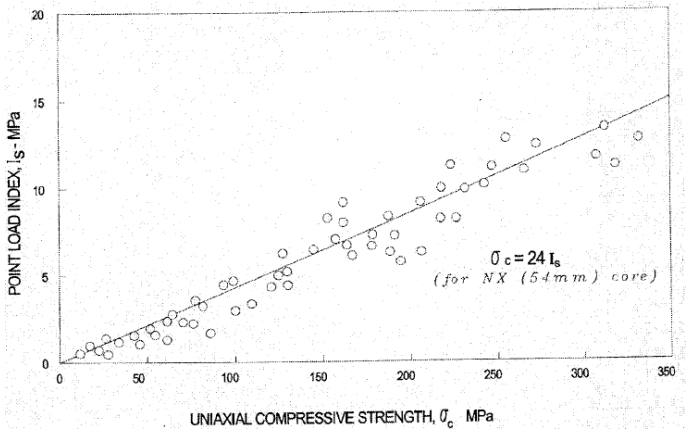


Figure 7 – Relationship between Point Load Strength Index and Uniaxial Compressive Strength

Remark: 1 MPa = 10.2 kg/cm^2 = 145 lb/in^2 (ISRM Suggested Methods)

TABLE 1 Generalized Value of "C"[A]

Core Size, mm	Value of "C" (Generalized)
20	17.5
30	19
40	21
50	23
54	24
60	24.5

[A] From ISRM Suggested Methods.[3]

Table 1 - Generalized Value of C

End of Procedure